Blinkit · DVA-capstone

The Unplanned Basket.

How quick-commerce is converting India's weekly grocery list into a daily stream of impulse spending — quantified across 13,000 SKUs and 10 cities.

SECTOR

Retail · Quick-Commerce

TEAM

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INSTITUTE

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SUBMISSION

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CONTEXT & PROBLEM

Quick-commerce shrunk the journey from craving to checkout. Nobody has priced what that's worth.

India's Q-commerce GMV crossed **₹64,000 Cr in FY25.** Blinkit, Zepto and Instamart compress need-to-purchase from days to seconds — structurally encouraging small-basket, unplanned spending.

The category is profitable per unit but margin-fragile. Every rupee of discount on a non-incremental SKU directly erodes contribution.

CORE QUESTION

What share of Blinkit's sales is impulse-driven — and how much more are consumers paying per gram for the convenience of small-pack, quick delivery?

DECISION SUPPORTED

Reallocate assortment investment toward high-impulse categories, reprice small packs against the empirical premium, and redirect promotional spend away from discounts that don't drive incremental volume.

DATA & PIPELINE

13,000 SKUs · 10 cities · ₹50.9 Cr revenue — end-to-end Python.

ROWS	COLUMNS	CITIES	CATEGORIES
13,000	**25**	**10**	**8**
Product-level snapshot	Raw + 10 engineered	Metros + tier-1	Bakery to Household

PYTHON ETL · 5 NOTEBOOKS

01 **Extraction** — profile 13K × 25 schema

02 **Cleaning** — recode 6,544 NaN offers; types

03 **EDA** — trend, distribution, correlation

04 **Statistical tests** — 5 hypotheses + OLS

05 **Tableau prep** — final_dataset.csv

ENGINEERED FEATURES DRIVING THE ANALYSIS

FEATURE	DEFINITION
`impulse_score`	0.30·sm + 0.25·per + 0.25·cat + 0.20·off
`basket_type`	Impulse ≥ 0.6 · Mixed · Planned ≤ 0.4
`price_per_100g`	final_price ÷ (weight_g ÷ 100)
`weight_bucket`	small · mid · bulk

KPI FRAMEWORK

Six metrics — every one of them ties to a category, pricing or promotional decision.

KPI · 01

Impulse Revenue Share

`Σ rev (Impulse) ÷ Σ rev × 100`

Sizes the impulse layer. Headline metric for the category & pricing teams.

KPI · 02

Convenience Premium

`median ₹/100g (small) ÷ median ₹/100g (bulk)`

The "tap tax" — the per-gram price of buying convenience.

KPI · 03

Impulse Score

`0.30·sm + 0.25·per + 0.25·cat + 0.20·off`

Transparent SKU-level score that drives basket bucketing.

KPI · 04

City Impulse Index

`city_mean ÷ global_mean`

Geographic comparison without size effects. Tests the case for city-level customisation.

KPI · 05

Offer Lift Ratio

`mean qty (offer) ÷ mean qty (no offer)`

Asks the only question that matters about promos.

KPI · 06

Top-20% SKU Concentration

`cum rev (top 20%) ÷ total rev`

Pareto check — informs assortment-pruning risk.

ALL KPIS COMPUTED IN 05_FINAL_LOAD_PREP.IPYNB

PART · 02

Findings.

Five questions, answered statistically. The headline is not in the volume — it is in the per-gram price.

INSIGHT 01 · REVENUE MIX

21.6% of revenue runs through impulse SKUs — almost half the catalogue carries an impulse signal.

TOTAL REVENUE	IMPULSE SHARE	MIXED SHARE	TOP 20% SKUS
₹509M	21.64%	27.14%	60.3%
13,000 SKUs · 10 cities	₹110.1M of unplanned revenue	₹138.2M with impulse signal	of revenue · Pareto partial

BASKET-TYPE REVENUE MIX

Planned · 5,117 SKUs	51.22%
Mixed · 4,139 SKUs	27.14%
Impulse · 3,744 SKUs	21.64%

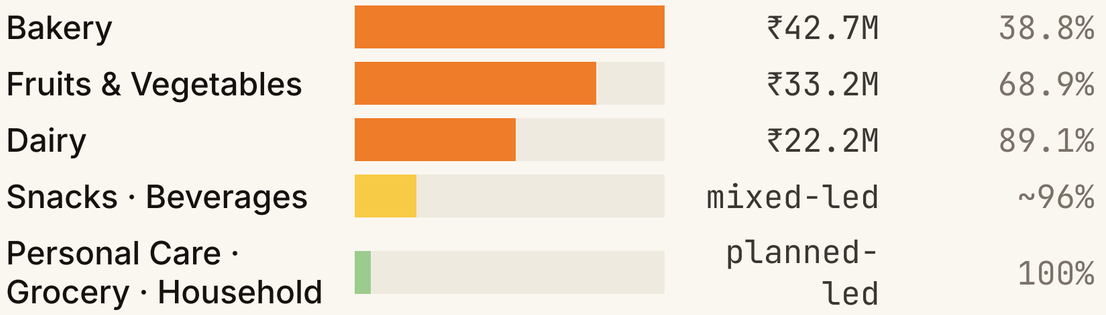
SO WHAT

The impulse layer is real. ₹110 M of revenue runs through unplanned-style SKUs. Including Mixed, **48.8%** of revenue carries an impulse signal.

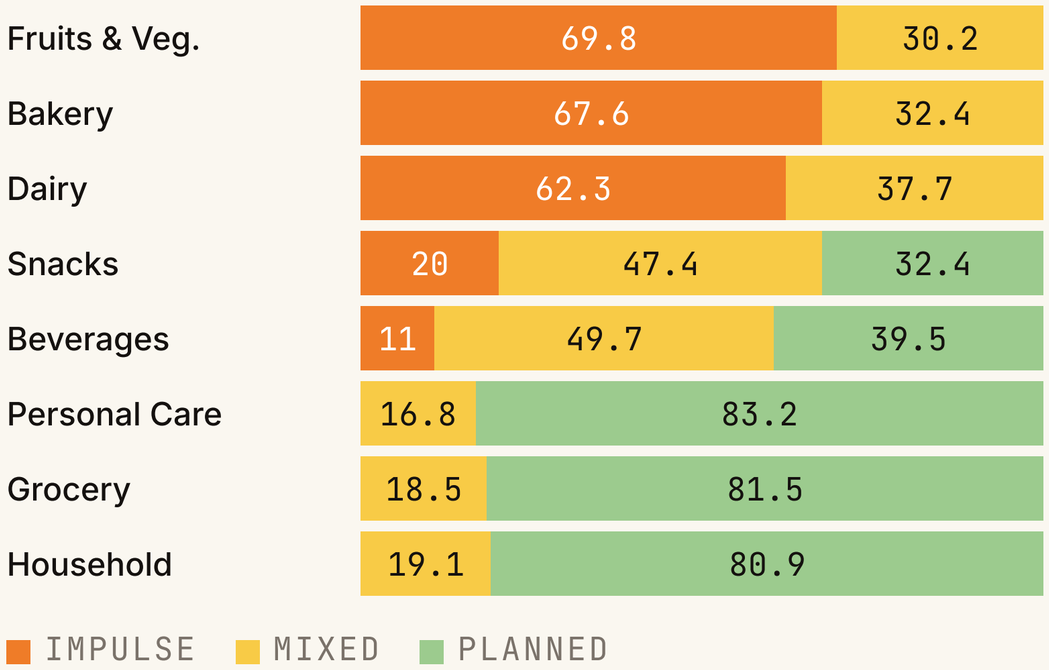
INSIGHT 02 · WHERE IMPULSE LIVES

Three categories own 89% of impulse revenue. The rest is structurally planned.

IMPULSE REVENUE PARETO



CATEGORY × BASKET COMPOSITION

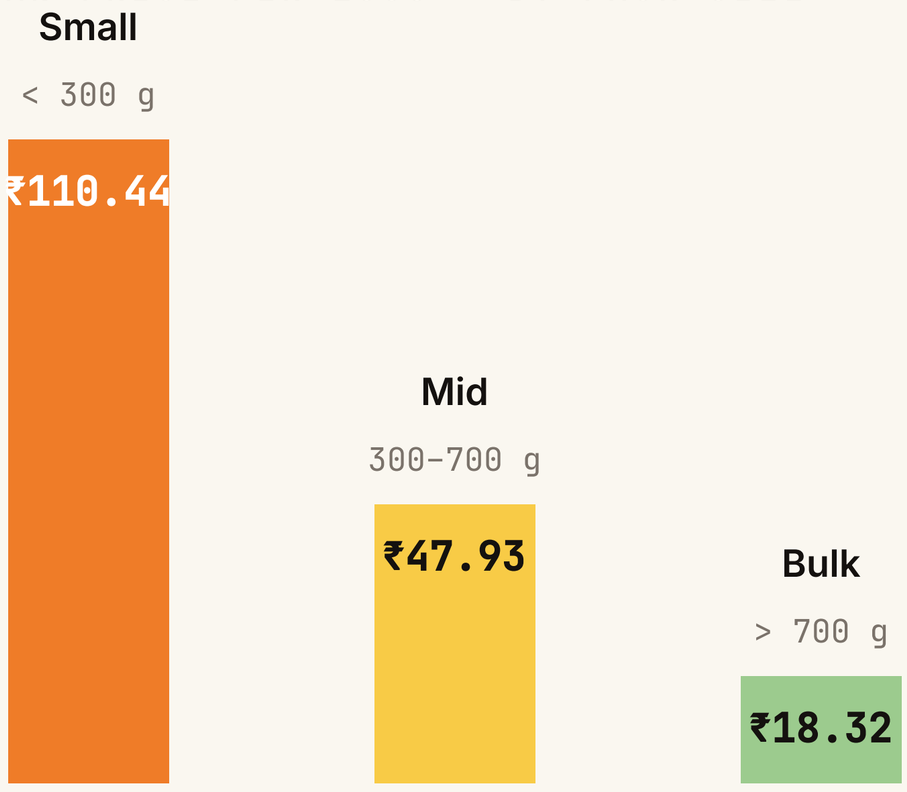


Short-shelf-life categories are the impulse core. Personal Care, Grocery and Household are structurally planned.

INSIGHT 03 · THE TAP TAX

Small packs cost 6× more per gram than bulk. This is the single strongest signal in the data.

MEDIAN PRICE PER 100G · BY PACK SIZE



PACK PREMIUM

6.0×

Small ₹110/100g vs Bulk ₹18/100g

STATISTICAL TEST · WELCH T

t = 49.40

p ≈ 0.000 — highly significant

IMPLICATION

₹/100g, not absolute ₹.

INSIGHT 04 · WHAT'S REAL

Five hypotheses tested. Only one returned a meaningful signal — and it isn't the one anyone expected.

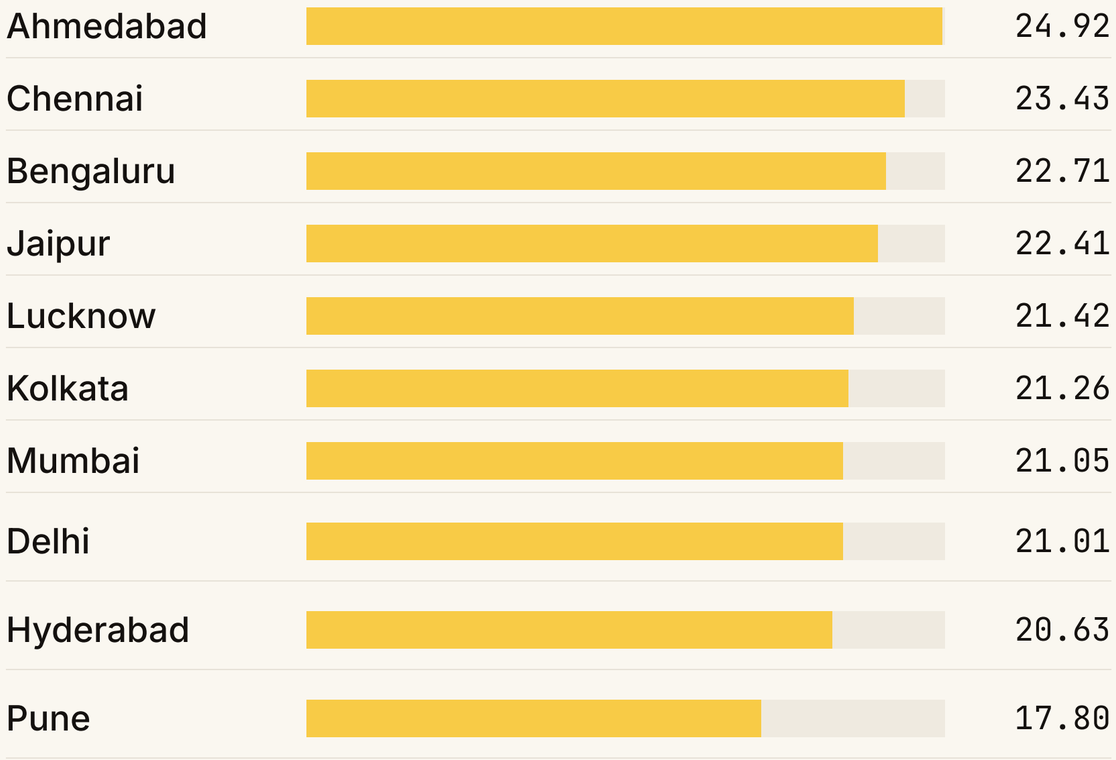
#	Hypothesis	Test	Statistic · p-value	Verdict
H1	Impulse SKUs sell more units than Planned	Mann-Whitney U	U=9.6M · p=0.804	NOT SIGNIFICANT
H2	Offers increase sold quantity	ANOVA	F=0.665 · p=0.415	NOT SIGNIFICANT
H3	Cities differ in impulse propensity	Kruskal-Wallis	H=9.13 · p=0.426	NOT SIGNIFICANT
H4	Small packs cost more per 100g	Welch t-test	t=49.40 · p≈0.000	HIGHLY SIGNIFICANT
H5	Top 20% of SKUs drive 80% of revenue	Pareto cumulative	share = 60.3%	PARTIALLY HOLDS

OLS R^2 = 0.000 · Impulse manifests through price premium, not volume uplift.

INSIGHT 05 · GEOGRAPHY & PROMOS

Geography is uniform. Discounts don't drive volume.

IMPULSE SHARE % BY CITY · MEAN 21.66%



Top vs bottom spread = 7.1pp · Kruskal-Wallis p = 0.43

OFFER LIFT RATIO

1.00×

161 vs 165 mean units · ANOVA p = 0.41

DISCOUNT → VOLUME R^2

0.000

No linear relationship · 0–30% range

OPERATIONAL OUTLIER

Lucknow & Jaipur — delivery delays at 31% / 28%

Impulse propensity is uniform; delivery operations are not. Localise opex, not category strategy.

SO WHAT

Cap discount depth at 10% on impulse SKUs.

RECOMMENDATIONS

Five moves. Each one anchored to a finding, sized to the dataset.

#	Insight → Action	What changes	Expected impact
R1	**Re-weight assortment toward Bakery, F&V, Dairy.**	These three categories drive 89% of impulse revenue. Concentrate in-app shelf, depth and cold-chain SLA.	**+₹5–10 Cr** +1–2pp impulse share
R2	**Expand small-pack SKUs in Snacks & Beverages.**	The 6× per-gram premium is un-monetised in mid-impulse categories. Launch small packs to capture it.	**+₹4 Cr** +50–100bps margin
R3	**Cap discount depth at 10% on impulse SKUs.**	ANOVA p=0.41, R^2=0.00 against demand. Redirect spend to placement, push and banner visibility.	**+₹2 Cr** profit at flat volume
R4	**Run one national impulse playbook.**	City variance not significant (p=0.43). Localise only delivery ops where Lucknow/Jaipur lag.	**Lower opex** faster rollout
R5	**Adopt demand_index as the reorder primary input.**	demand_index ↔ sold_qty correlation = 0.91. Already in-schema; needs only operational adoption.	**-200 bps** stockout target

IMPACT

The next ₹5–10 Cr of margin lives in pack-size economics — not in deeper discounts.

₹13Cr+

order-of-magnitude annualised value across R1–R5, anchored to **₹509M revenue**.

COST OF INACTION

Margin defensibility comes from price architecture, not delivery speed.

LEVER	FROM	TO	VALUE
R1 · Assortment	21.6%	22.6%	+₹5 Cr
R2 · Small packs	Sn 20 / Bv 11	+5pp each	+₹4 Cr
R3 · Discount cap	0–30%	≤10%	+₹2 Cr profit
R4 · National play	10-city	1 national	opex saved
R5 · demand_index	current	-200 bps	+₹1–2 Cr

LIMITATIONS & NEXT STEPS

What this study can't yet answer.

LIMITATIONS

- **L1** **Snapshot, not transactions.** No time-series.
- **L2** **Inferred impulse.** Rule-based, not behavioural.
- **L3** **50% missing offer_type.** H2 may be conservative.
- **L4** **Analyst-set weights.** 0.30/0.25/0.25/0.20.
- **L5** **No causal claims.** Correlations only.

NEXT STEPS

- **N1** **Transactional enrichment** · order-level log.
- **N2** **Sensitivity grid** on impulse weights.
- **N3** **Tree-based demand model** beyond linear.
- **N4** **Dark-store catchment geo** — sub-city.
- **N5** **A/B test on R3** to causally validate.

CLOSING

Quick-commerce was supposed to be all impulse. The data says it is more subtle than that.

21.6% of revenue is impulse — concentrated in three categories, monetised through a 6× per-gram premium on small packs. Promotions barely matter. The next ₹5–10 Cr of margin lives in pack-size economics.

github · Arya756/DVA_Capstone2_C_G19 tableau · Blinkit_Analysis_G19.twbx team · DVA-C-G19